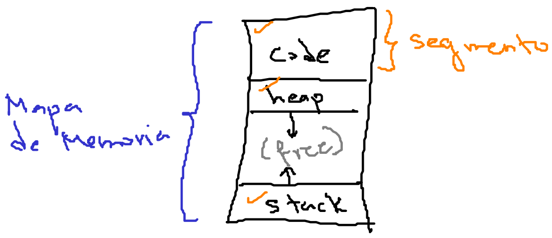
24/03/2026 - Sistemas Operativos (Ude@)

Segmentación

Conceptos importantes:

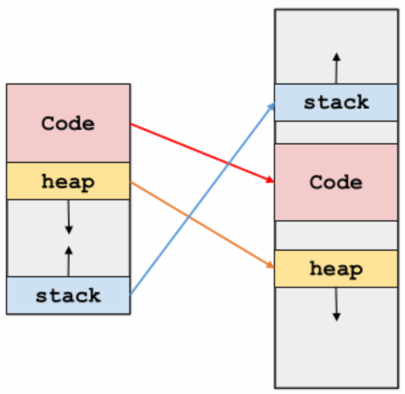
1. Segmento

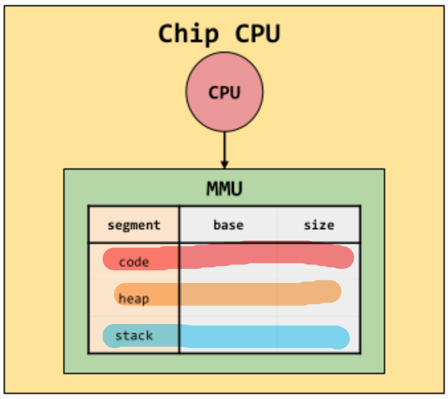


2. Los segmentos están desacoplados → Cada segmento {code, heap, stack} tiene sus propios base & bound

segmentación

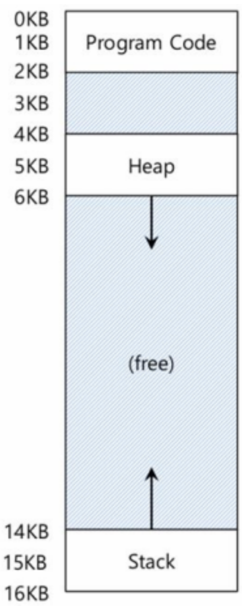
bound → size





3. Traducción de direcciones.

→ Depende del segmento



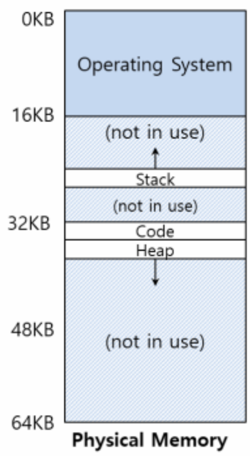
① Check Limits:

$0 \leq \text{offset}[seg] \leq \text{size}[seg]$

VA → AT → PA

AT

segment	base	size
code	32K	2K
heap	34K	2K
stack	28K	2K

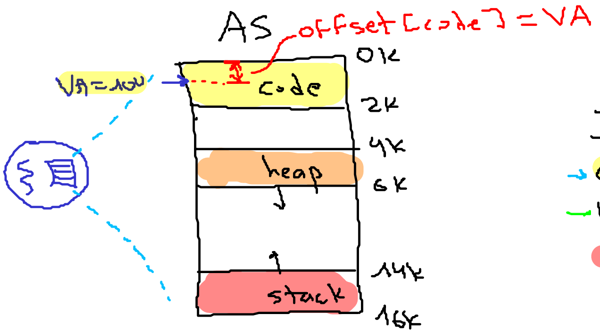


② Obtener la PA

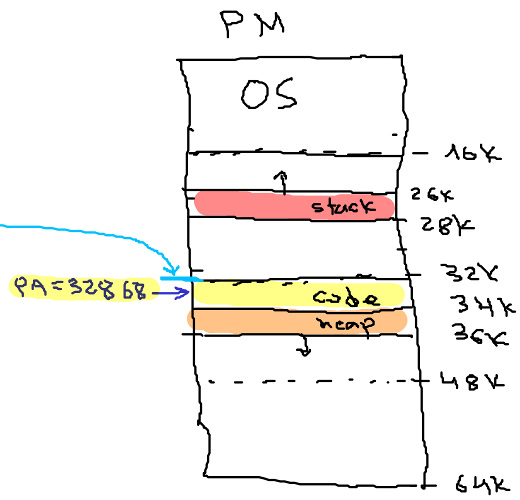
$PA = \text{base}[seg] + \text{offset}[seg]$

la expresión cambia según el segmento

i. VA este en el segmento de code



seg	base	size
code	32K	2K
heap	34K	2K
stack	28K	2K



size(AS) = 16K

size(PM) = 64K

Si VA = 100 → PA = ?

→ offset[code] = VA = 100

{ base = 32K
size = 2K

(1) $0 \leq \text{offset} < \text{size}$

$0 \leq 100 < 2K$ ✓

(2) $PA = \text{base} + \text{offset}$

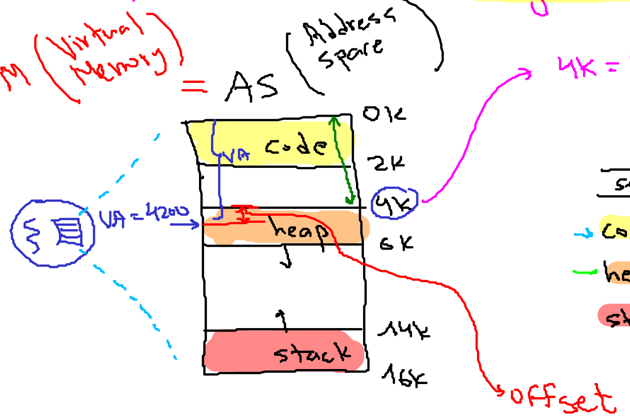
$PA = 32K + 100 = 32(1024) + 100 = 32868$

ii. VA este en el segmento heap

VM (Virtual Memory) = AS (Address Space)

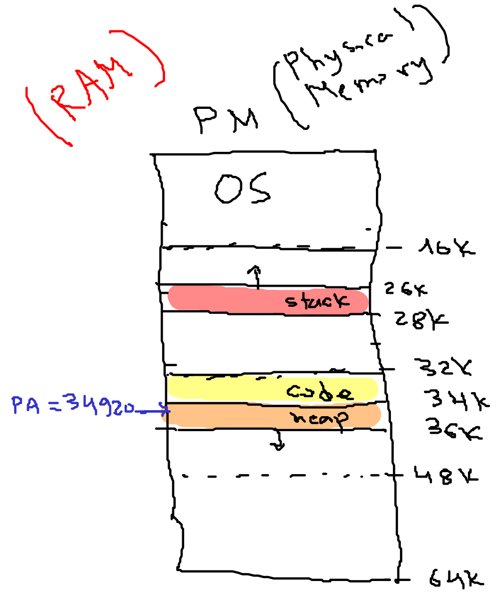
(RAM) PM (Physical Memory)

$4K = 4(1024) = 4096$



fisico

seg	base	size
code	32K	2K
heap	34K	2K
stack	28K	2K



size(AS) = 16K

size(PM) = 64K

Si VA = 4200 → PA = ?

{ size = 2K ✓
base = 34K ✓

→ Heap: offset[heap] = VA − 4K = 4200 − 4096 = 104

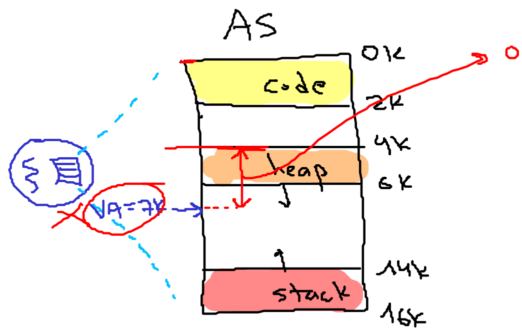
(1) $0 \leq \text{offset} < \text{size}$

$0 \leq 104 < 2K \rightarrow 0 \leq 104 < 2048$ ✓

(2) $PA = \text{base} + \text{offset}$

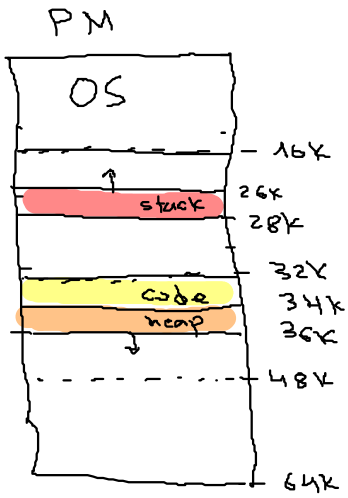
$PA = 34K + 104 = 34(1024) + 104 = 34920$

iii. VA es invalida.



size(AS)=16K

seg	base	size
code	32K	2K
heap	34K	2K
stack	28K	2K



size(PM)=64K

Si VA = 7K ⟶ PA = ?

heap: { base= 34K, size = 2K

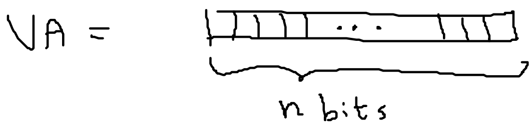
offset = VA − 4K = 7K − 4K = 3K = 3(1024) = 3072

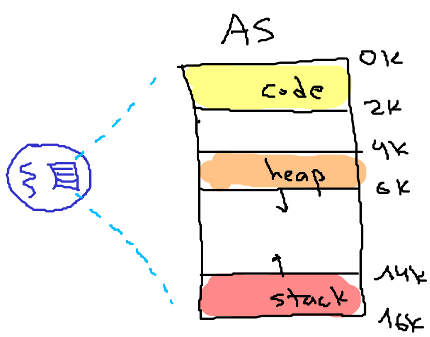
(1) 0 < offset ≤ size

0 < 3K ≤ 2K (x) Invalido

seg fault (No hay PA asociada)

Como es el Formato de direcciones para una VA (Virtual Address)





size(AS)=16K

Si size(AS) = 16K ⟶ n = ?

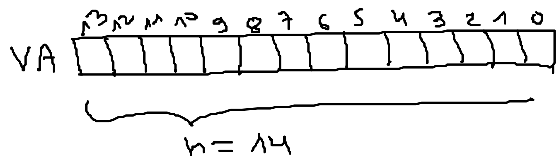
2^n = size(AS)

$2^n = 16K = 2^4(2^{10})$

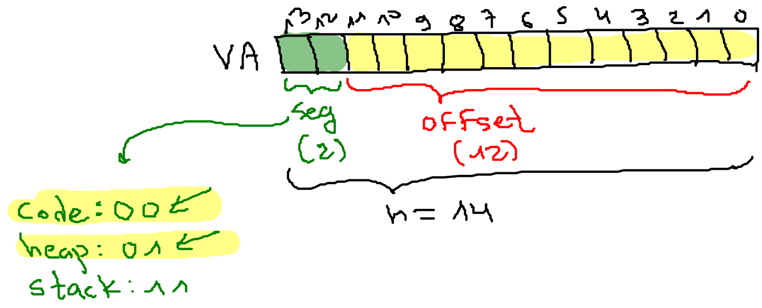
$2^n = 2^{14}$

$\boxed{n = 14}$

Direccion Logica



Rango: $[0, 2^n - 1] = [0, 2^{14} - 1] = [0, 16383]$

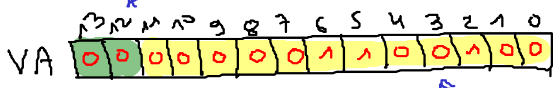


¿Cual es la representacion de las direcciones Logicas anteriores?

① Code: VA = 100 ⟶ offset = VA = 100 = 0x64 = $\underbrace{0110}_{6}\,\underbrace{0100}_{4}$

seg = 00

= 00000001100100

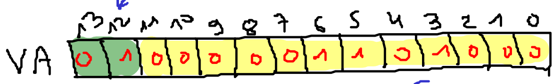


② Heap: VA = 4200 ⟶ offset = VA − 4K = 4200 − 4096 = 104

= 104 = 0x68 = $\underbrace{0110}_{6}\,\underbrace{1000}_{8}$

seg = 01

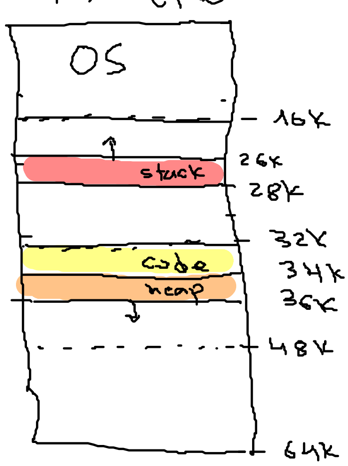
= 00000110 1000



③ Direccion invalida: VA = 7K (heap) ✗ No se puede Mapear

Como es el Formato para las direcciones Fisicas

PM (Physics Memory)



size(PM) = 64K

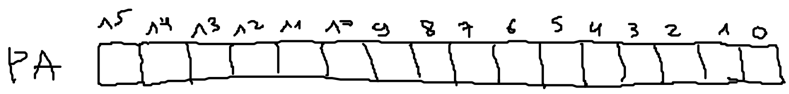
¿Si size(PM) = 64K cuantos bits necesito?

$$2^n = size(PM)$$

$$2^n = 64K = 2^6(2^{10})$$

$$2^n = 2^{16}$$

$$n = 16$$



Rango de direcciones: $[0, 2^n - 1]$

$[0, 2^{16} - 1] = [0, 65535]$

En el ejemplo:

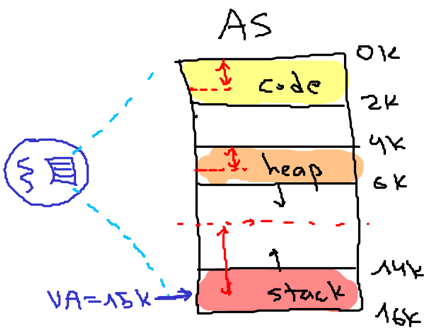
a. $VA = 100 = 0x64 = [00 \vdots 0000\,0110\,0100]$

$(n=16)$ $PA = 32868 = 0x8064 = [\underbrace{1000}_{8}\ \underbrace{0000}_{0}\ \underbrace{0110}_{6}\ \underbrace{0100}_{4}]$

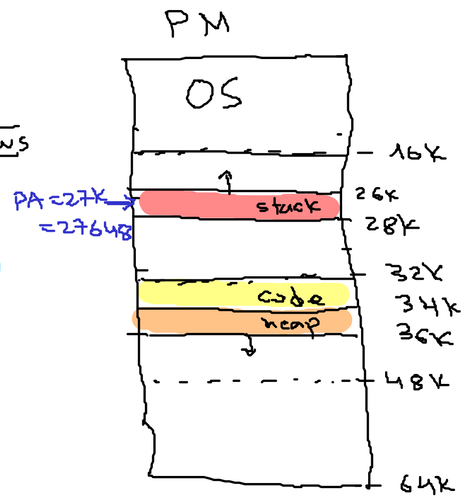
b. $VA = 104 = 0x68 = [01 \vdots 0000\,0110\,1000]$

$(n=16)$ $PA = 34920 = 0x8868 = [\underbrace{1000}_{8}\ \underbrace{1000}_{8}\ \underbrace{0110}_{6}\ \underbrace{0100}_{8}]$

(4) Stack: $VA = 15K$



	seg	base	size	grows
00 →	code	32K	2K	1
01 →	heap	34K	2K	1
11 →	stack	28K	2K	0



size(AS) = 16K

$VA = 15K \longrightarrow$ seg = stack → base = 28K, size = bounds = 2K

size(PM) = 64K

a. Formato de direcciones $(n=14)$

$VA = 15K = 15(2^{10}) = 15(1024) = 15360 = 0x3C00$

$VA = 0x3C00 = [\underbrace{\cancel{00}11}_{3}\ \underbrace{1100}_{C}\ \underbrace{0000}_{0}\ \underbrace{0000}_{0}]$ $(n=14)$

$VA = [11 \vdots 1100\,0000\,0000]$

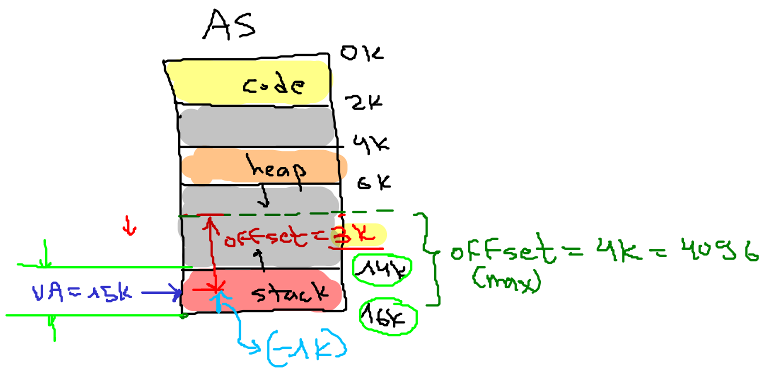
11 → stack

offset: $0xC00 = 3072 \times \frac{1K}{1024} = 3K$

$2^{12} = 2^2\, 2^{10} = 4K$

$n(\text{offset}) = 12 \rightarrow [0, 2^{12} - 1] = [0, 4095]$

→ 4096 posibles offset → offset(max) = 4K



Entonces como obtengo la direccion fisica (PA)?

VA = 15K = [11 | 1100 0000 0000]

base = 28K
bounds = size = 2K

offset = 3K
offset(max) = 4K

offset (stack) = offset − offset (max)

offset (stack) = 3K − 4K

offset (stack) = −1K

① check Limits ✓

② Calculo de PA: PA = base + offset

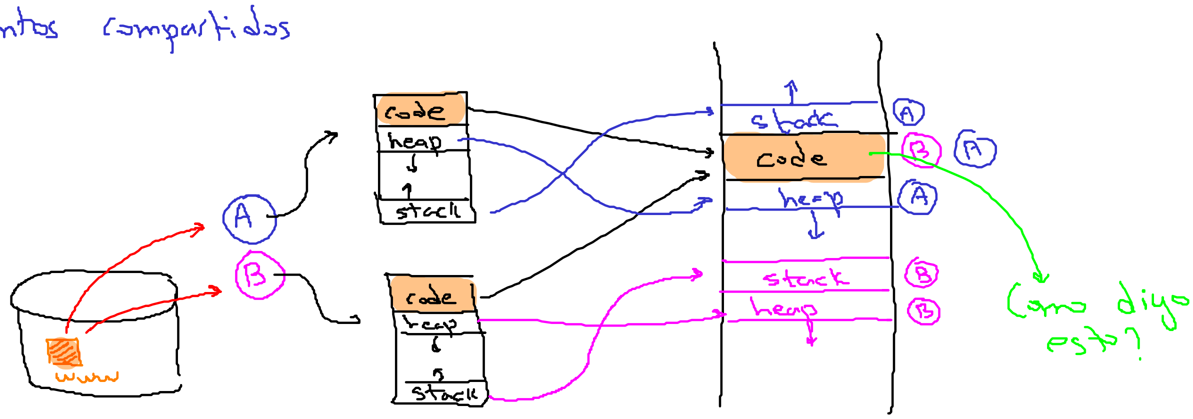
PA = 28K + (−1) = 27K

PA = 27 (1024) = 27648 = 0x 6C00

Luego PA (n = 16 bits):

$$PA = \underbrace{0110}_{6}\ \underbrace{1100}_{C}\ \underbrace{0000}_{0}\ \underbrace{0000}_{0}$$

Segmentos compartidos



R Read W Write X Execute

segment	base	size (max 4K)	Grows Positive?	Protección
code(00)	32K	2K	1	Read-Execute (R-X)
heap(01)	34K	2K	1	Read-Write (RW-)
stack(11)	28K	2K	0	Read-Write (RW-)

Segment Register Values (with Protection)